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Entropy driving highly selective CO_2 separation in nanoconfined ionic liquids

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ABSTRACT

Nowadays, the global greenhouse effect has led to the imminent development of CO_2 capture, separation, and storage technologies. Hybrid membranes with nanoconfined ionic liquids (ILs) show great potential for CO_2 separation, but the intrinsic mechanism is still obscure. Herein, the thermodynamical properties and solvating processes of CO_2 and CH_4 in ILs confined in graphene oxide were studied via performing massive molecular dynamics simulations. It was first identified that selectivity rises from 25.01 to 149.20 as the interlayer distance decreases from 3.00 to 1.50 nm, showing an ultrahigh separating selectivity. Interestingly, the solubility of CO_2 in confined ILs increases by almost two orders of magnitude compared with that in bulk ILs, which is far larger than CH_4 in confined ILs. The high solubility mainly originates from the fact that the confined ILs can induce the structure rearrangement and provide abundant CO_2 adsorbing sites, raising the configurational entropy of CO_2 in the confined ILs, and further driving the high separation selectivity of CO_2 over CH_4. Finally, quantitative relations between solubility, diffusion capacity, permeability, selectivity, and structural entropy of gas in confined ILs are constructed, which are meaningful for the theoretical understanding, rational design, and applications of highly efficient and low-cost separation of CO_2.

1. Introduction

Pushed by the rapid growth of the global economy and energy demand, the atmospheric concentration of carbon dioxide (CO_2) rises significantly, contributing to climate change [1,2]. Meanwhile, CO_2 is an undeniable source of chemicals and fuels [3,4]. Thus, developing CO_2 capture, separation, and storage technologies are vital, witnessing growing academic interests [5,6]. Recently, abundant separation technologies [7–12] gradually matured, as the easy processability and high scale-up capability, membrane separation [10,11] exhibit great potential. Two-dimensional (2D) materials, such as graphene oxide (GO) and molybdenum disulfide (MoS_2), is feasible to be designed with precise channel dimensions and target chemical functions to achieve extraordinary selectivity [13,14], opening a new era of membranes. However, the membranes prepared only by 2D materials have limited performance [11,15,16]. Considering the high stability and high affinity for CO_2 [6,17–22], ionic liquids (ILs) can serve as the ideal fillers for the membrane to overcome the challenge of membrane selectivity. Such 2D materials-supported ILs membranes (2D-SILMs), consisting of many regulatable nanochannels, are promising candidates for CO_2 separation, theoretically providing the ultimate penetration performance [23,24] and even breaking the permeability-selectivity competitive bottleneck.

Recently, extensive works have been implemented to clarify the structures, properties, and transportation of CO_2 and CH_4 in 2D-SILMs, where the high separation of CO_2 over CH_4 is meaningful for reducing the separation energy consumption of industrial mixtures such as syngas. Gu et al., [17] experimentally demonstrated that imidazolium cations could be coupled with GO sheets by π-π, cation-π, and electrostatic interactions, further effectively suppressing the interlayer distance (d) expansion of GO-SILMs, which can show high stability in long-time aqueous immersion. Moreover, via a simple dip-coating technique, Ying et al., [25] prepared a high-performance GO-SILM by confining 1-butyl-3-methylimidazolium tetrafluoroborate ($BmimBF_4$) into a GO membrane with d = 1.94 nm, where the CO_2 over methane (CH_4)

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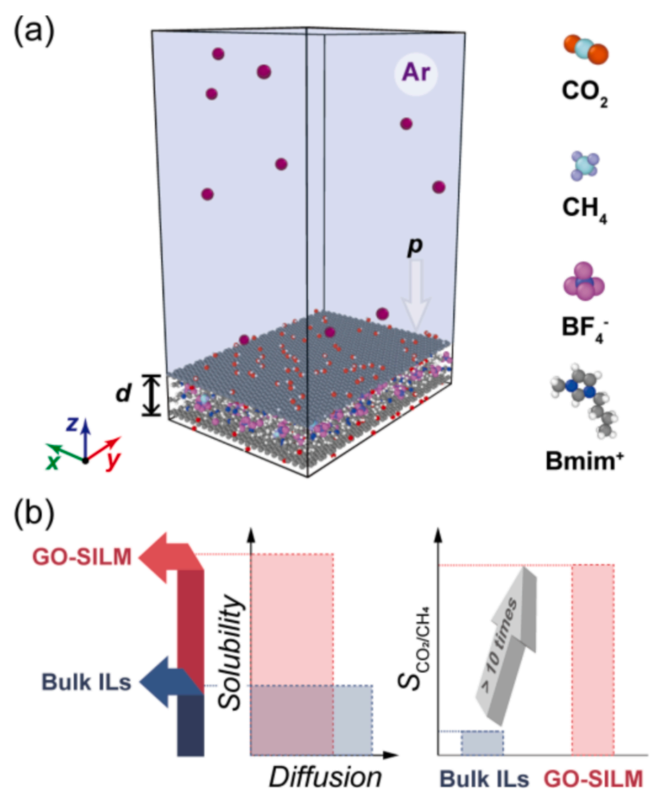

Fig. 1. Setup of the simulation. (a) Left: the atomic structure of the composite system, where gray, white, and red colors represent carbon, hydrogen, and oxygen atoms of a BLGO, respectively. The lavender shaded area is filled with Ar atoms (violet color). Right: atomic structures of $Bmim^+$, BF_4^-, CO_2, and CH_4, where gray, white, blue, violet, and magenta colors represent carbon, hydrogen, nitrogen, boron, and fluorine atoms of $BmimBF_4$, cyan, orange, and iceblue colors represent carbon, oxygen, and hydrogen atoms of gases, respectively. (b) The diffusion, the solubility of gases, and $S_{CO2/CH4}$ in the GO-SILM and bulk ILs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

permeability selectivity ($S_{CO2/CH4}$) can reach up to 234, and the permeance of CO_2 is 68.5 GPU. As a comparison, the $S_{CO2/CH4}$ of dry and wet GO membranes [26] are only 8.8 and 32, respectively. Even more astonishing, by applying an external electric field [27], the $S_{CO2/CH4}$ of the GO-SILM with $d = 2.11$ nm is dramatically increased to 262, and the CO_2 permeance is raised to 260 GPU. Furthermore, the mica [28], MoS_2 [29], and tungsten disulfide [30] supported ILs membrane was constructed by immobilizing $BmimBF_4$ into 2D nanosheets, where the CO_2 permeance can up to 80, 47.88, and 47.3 GPU, and the values of $S_{CO2/CH4}$ are 28.6, 43.52 and 68.81, respectively. Karunakaran et al., [31] also produced a GO-SILM by confining 1-ethyl-3-methylimidazolium acetate inside a GO membrane, where the $S_{CO2/CH4}$ is 39. Except for conventional ILs, Wan et al., [32] immobilized magnetic ILs into boron nitride to form a 2D-SILM with a $S_{CO2/CH4}$ of 38, and the CO_2 permeance is 20.3 GPU. Hence, confined ILs can indeed boost the selectivity and permeability of CO_2 synergistically, which depends greatly on the properties of ILs-solid interfaces.

Such a high $S_{CO2/CH4}$ of the GO-SILM mainly stems from three primary factors. First, $BmimBF_4$ is an IL with delight CO_2 solubility and high $S_{CO2/CH4}$ [18,19], sealing the large, non-selective defects and wrinkles of the GO-SILM [25]. Second, solubility selectivity should dominate $S_{CO2/CH4}$ in the GO-SILM [6]. Third, d of the GO-SILM plays a crucial role in the fast and selective transport of species separated [33,34]. The excellent selectivity of GO-SILMs suggests that GO-SILMs should be a preferred candidate in the efficient separation of CO_2. However, beyond qualitative understandings of the three primary factors, quantitative characterization is very scarce so far. For instance, the relationships within $S_{CO2/CH4}$, permeability, solubility, diffusion capacity, structure, and thermodynamic properties of CO_2 and CH_4 in GO-SILMs with varying d are still not clear. Such confusion restricts the theoretical understanding, rational design, and applications of highly efficient and low-cost separation of CO_2.

Herein, massive molecular dynamics (MD) simulations were performed to reveal the structure, thermodynamic properties, and transportation of CO_2 and CH_4 in $BmimBF_4$ confined in bilayer GO nanochannels (BLGOs). We aimed to assess the role of entropy on $S_{CO2/CH4}$, gaining deeper insights into how d affects the thermodynamic properties of confined gases and how it further determines the structure, diffusion capacity, solubility, permeability, and selectivity of confined gases. The mechanism that entropy driving highly selective CO_2 separation in GO-SILMs is further revealed. By comparing the control range of temperature (T), pressure (p), the concentration of the hydroxyl group (c_{OH}), and d on $S_{CO2/CH4}$, the effective way to improve $S_{CO2/CH4}$ is proposed. Finally, the role of BLGO and ILs in the dissolution of gases is quantified, providing more support for improving the solubility of gases in 2D nanosheets-SILMs. Such quantitative characterizations are meaningful for the theoretical understanding, rational design, and applications of highly efficient and low-cost separation of CO_2.

2. Computational Details

2.1. Atomic model of the GO-SILM

As shown in Fig. 1**a**, the IL considered in this work is $BmimBF_4$, confined in a BLGO and further forming a GO-SILM. Hydroxyl groups of a BLGO are randomly distributed on two sides of the graphene nanosheet, and the ratio of hydroxyl groups to carbon atoms is $c_{OH} = n_{OH}/n_C = 10\%$. The distance between two GO nanosheets, d, varies from 1.25 to 4.00 nm, leading to a series of GO-SILMs. To apply external pressure to the system, the gaseous argon (Ar) atoms are randomly placed above the upper GO wall. The in-plane (x-y) size of the BLGO is 5.40×7.66 nm^2, and a periodic boundary condition (PBC) is applied in the x-y plane while an open boundary is used along the z-direction. CO_2 or CH_4 molecules are further confined in GO-SILMs to explore their transportation, as referenced to gases in bulk ILs (Fig. 1**b**).

2.2. Force field of the GO-SILM

The parameters of the bond, angle, dihedral, van der Waals interactions, and electrostatic interactions of $BmimBF_4$ and GO were described by the all-atom optimized potentials for liquid simulations (OPLS-AA) force field [35,36], which has been successfully used to obtain the structures and properties of both $BmimBF_4$ and GO [37]. Meanwhile, carbon atoms in BLGOs were fixed to maintain the planar sheets during the entire simulation. Given the reported literature [38,39], CO_2 was modeled using the transferable potentials for the phase equilibria-explicit hydrogen model [40], and CH_4 was described by the OPLS-AA force field [36]. The interactions between gases, $BmimBF_4$, and GO can be divided into electrostatic and van der Waals terms. The former one, the long-range coulombic interaction, was computed using the particle–particle–particle-mesh (PPPM) algorithm [41], while the latter was represented by using the 12–6 Lennard-Jones (LJ) potential, truncated at 1.2 nm, and geometric combining rules were utilized. To apply the same parameter for intra- and intermolecular interactions, the 1, 4-intramolecular interactions were reduced by a factor of 2 [35]. The SHAKE algorithm [42] was applied to O–H bonds to reduce high-frequency vibrations.

2.3. Simulation details

All MD simulations in this work were carried out via the large-scale atomic/molecular massively parallel simulator (LAMMPS) [43]. The time step was 2.0 fs, and the pair number of $BmimBF_4$ was defined as N_{IL}

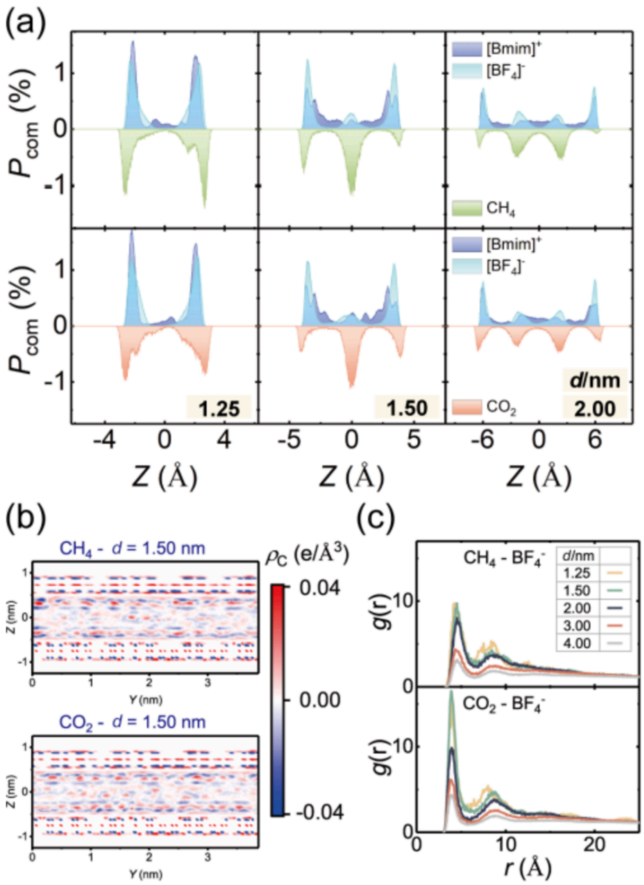

Fig. 2. Distribution of gases in GO-SILMs. (a) The spatial distribution probability of the centers of masses (P_{com}) for $Bmim^+$, BF_4^-, CO_2, and CH_4 along the *z*-direction. (b) The charge density (ρ_c) profile along the *y-z* plane for the composite system with $d = 1.5$ nm. (c) Radial distribution functions (RDFs) for CH_4-BF_4^- and CO_2-BF_4^- pairs, as functions of the distance (*r*).

(**Figure S1**). For the simulation system of self-diffusive coefficient (SDC), owing to the ensemble average [39], the number of gas molecules was $N_{gas} = 10\%N_{IL}$. Gases and $BmimBF_4$ were relaxed in the NVT ensemble with $T = 300$ K, controlled by using the Berendsen thermostat [44]. After a 15 ns-long MD simulation, the composite system was in equilibrium. MD simulations continue to run for an additional 5.0 ns to collect data for analyzing the structure, properties, and diffusion capacity of CO_2 and CH_4 in GO-SILMs. The entropy (*S*), enthalpy (*H*), and Helmholtz free energy (*A*) of confined gases were also calculated along equilibrated trajectories using the two-phase thermodynamics (2PT) free energy method [45]. Furthermore, the *S*, *H*, *A*, hydrogen bonds (HBs), and in-plane SDC (D_{xy}) were obtained by statistical averaging over five independent simulations.

In the simulation system of solvation free energy (ΔG_{sol}), $N_{gas} = 1$, due to the requirement of one solute molecule [38,39]. ΔG_{sol} is the free energy change of transferring an isolated gas molecule from an ideal gas phase ($state_0$) to a solution phase ($state_1$) at a given (*T*, *p*), playing a vital role in Henry's law constant (K_H) and solubility (*x*) [38]. In this work, we regarded a GO-SILM as the solvent. In $state_0$, the gas molecule only interacted with itself, while the gas molecule interacted with all solvent atoms in $state_1$. After the system was in equilibrium, MD simulations continued to run for an additional 30.0 ns to gather data for acquiring ΔG_{sol}. We implemented a reversible thermodynamic path by creating a sequence of alchemical intermediate states ($state_\lambda$) with the coupling parameter λ, where λ was adjusted linearly every 150 ps. Modifying LJ and coulombic interactions between the gas molecule and the solvent were straightforward by the soft-core potential (**Figure S2**) in LAMMPS [46]. The potential energy of $state_\lambda$ was defined as $U_\lambda = (1\text{-}\lambda)U_0 + \lambda U_1$, where U_0 and U_1 are potential energies at $state_0$ and $state_1$. Then, ΔG_{sol} of the gas molecule was calculated via the free energy perturbation (FEP) [47] method:

$$\Delta G_{sol} = \sum_{\lambda=0}^{1} \Delta G_\lambda = \sum_{i=0}^{n-1} \Delta_{\lambda_i}^{\lambda_{i+1}} G_i = -kT \sum_{i=0}^{n-1} ln \left\langle exp\left(-\frac{U(\lambda_{i+1}) - U(\lambda_i)}{kT} \right) \right\rangle_{\lambda_i} \quad (1)$$

where $\Delta\lambda = 0.005$, $n = 1/\Delta\lambda = 200$, *k* is the Boltzmann constant, ΔG_λ is the free energy difference at the specific λ.

3. Results and discussions

3.1. Structural feature of gases in GO-SILMs

The separating process and performance are tightly correlated with the structural features of the gases in GO-SILMs. First, the spatial distribution probability of the center of mass (P_{com}) for $Bmim^+$, BF_4^-, CO_2, and CH_4 along the *z*-direction is analyzed (Fig. 2**a&S3-4**). It clearly shows that compared with $Bmim^+$ and BF_4^-, gases are distributed closer to GO walls. Simultaneously, both CO_2 and CH_4 are distributed in the region occupied by BF_4^- rather than that occupied by $Bmim^+$, indicating the dominant role of BF_4^-. For CO_2 in the GO-SILM with $d = 2.00$ nm, gas molecules are mainly distributed in two regions: near-wall layers and the center of the BLGO. As *d* declines to 1.50 nm, two near-wall CO_2 layers are visible, while a strong CO_2 layer is located at the center. With *d* reducing to 1.25 nm, two near-wall CO_2 layers are still observed, while the strong CO_2 layer disappears. The charge density (ρ_c) profile along the *y-z* plane for the composite system with $d = 1.50$ nm is also displayed in Fig. 2**b**, showing the well-defined positive and negative charges near the GO walls, reflecting the strong electrostatic interaction. The radial distribution functions (RDFs) for CH_4-BF_4^- and CO_2-BF_4^- pairs are further calculated (Fig. 2**c**). As *d* grows, the RDF peak position for both CO_2 and CH_4 change little while the peak intensity decreases distinctly, indicating that confinements promote more gas molecules to enter the region occupied by BF_4^-. The RDF peak strength of CH_4-BF_4^- is always weaker than that of CO_2-BF_4^-, suggesting that BF_4^- possesses a higher affinity with CO_2.

The distribution difference between CO_2 and CH_4 mainly arises from the various HB interactions, including $O\text{-}H{\cdots}C_4$ and $O\text{-}H{\cdots}O_2$, where O, H, C_4, and O_2 represent oxygen and hydrogen atoms in the hydroxyl group, the carbon atom in CH_4, and oxygen atoms in CO_2, respectively. As shown in Fig. 3**a&S5**, combined distribution functions of angle ($A_{O\text{-}H{\cdots}C4}$ and $A_{O\text{-}H{\cdots}O2}$) with distance ($r_{H{\cdots}C4}$ and $r_{H{\cdots}O2}$) for different composite systems are summarized. Evidently, for CO_2 molecules in the GO-SILM with $d = 1.25$ nm, most $A_{O\text{-}H{\cdots}O2}$ concentrates at $\sim 90^\circ$, and the majority of $r_{H{\cdots}O2}$ focuses on 0.355 nm. For CH_4 molecules in the GO-SILM, possessing the tetrahedral structure, although a large portion of $A_{O\text{-}H{\cdots}C4}$ is scattered around $\sim 135^\circ$ and $\sim 105^\circ$, the majority of $r_{H{\cdots}C4}$ concentrates at 0.755 nm (**Figure S6**), indicating the slacker HB interaction between CH_4 and hydroxyl groups, which is in accordance with the higher interaction energy (E_{int}) between CH_4 and GO (**Figure S7**). Stunningly, with *d* going up, most $r_{H{\cdots}O2}$ still focuses on 0.355 nm, while only individual points in the region of $r_{H{\cdots}C4} < 0.50$ nm, demonstrating almost no HB formed between CH_4 and GO walls.

The average number of HBs per gas molecule (N_{HB}) is also tracked in Fig. 3**b**. As *d* rises, although the N_{HB} of both CO_2 and CH_4 decreases, the N_{HB} of CO_2 is at least 9 times compared to CH_4, echoing that the fraction of interfacial CO_2 is always greater than that of CH_4 (**Figure S4**). Physically, the decrease in N_{HB} arises from confined gas molecules migrating from near-wall layers to the center of BLGOs. The restriction from GO walls is weaker and weaker, allowing gases to exist in a more chaotic form and may achieve larger *S*. To detect the nature of confined CO_2 and CH_4, the typical vibrational spectrum, called the translational density of state (*TDOS*), is obtained, as shown in Fig. 3**c**. As *d* decreases

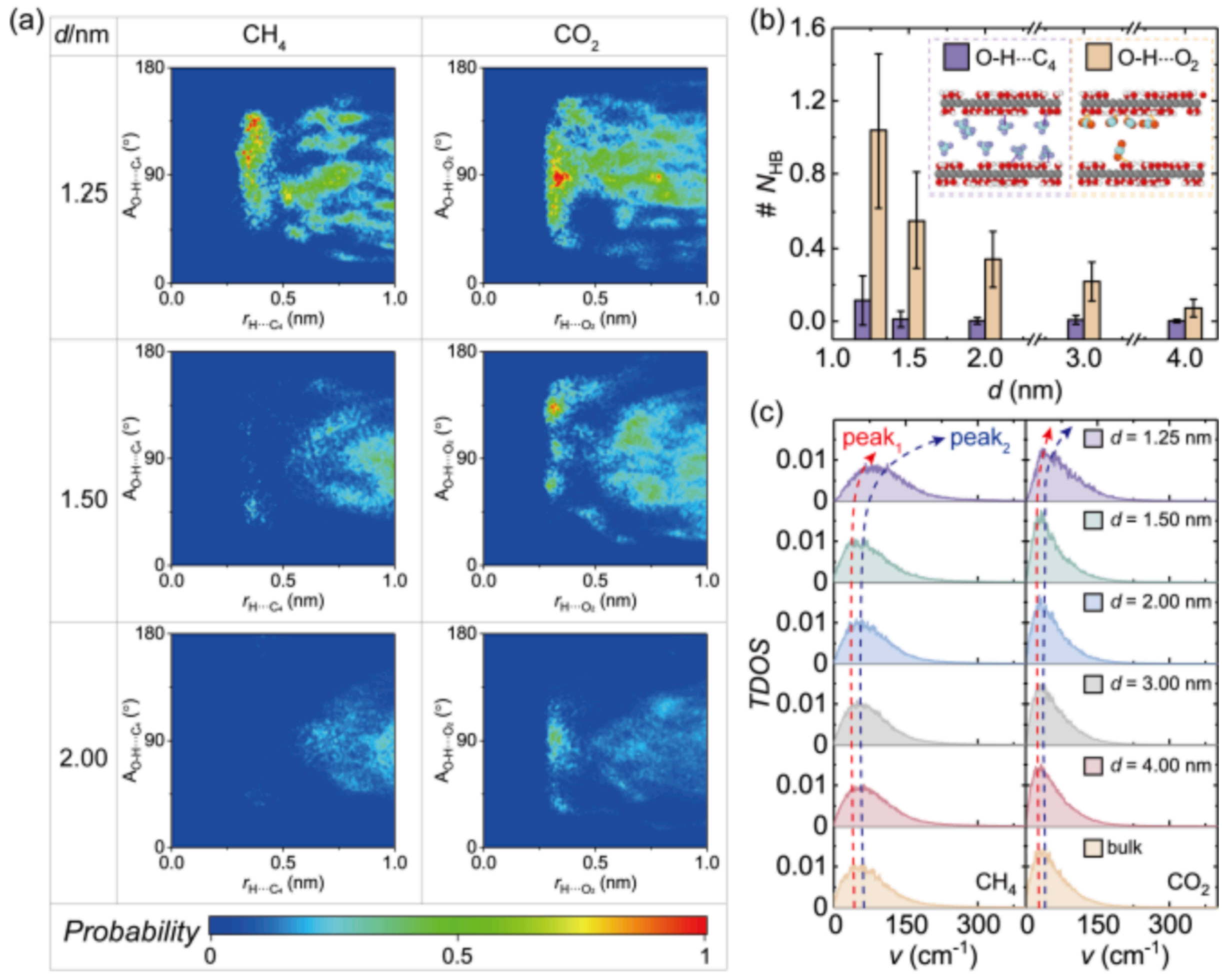

Fig. 3. HB networks and vibrational spectrums of gases in GO-SILMs. (a) Combined distributions of $A_{O\text{-}H\cdots C4}$ with $r_{H\cdots C4}$ and $A_{O\text{-}H\cdots O2}$ with $r_{H\cdots O2}$. (b) The average number of HBs per gas molecule (N_{HB}) changes with *d*. Insets are schematic diagrams of HBs between gas molecules with GO walls (O-H•••C_4 and O-H•••O_2). (c) The translational density of state (*TDOS*) of gases in the GO-SILM with different *d*, compared to gases in bulk ILs, respectively.

from 4.00 to 1.25 nm, the *TDOS* peaks of CO_2 and CH_4 show a clear blue shift, matching well with the evolution of N_{HB}-*d*. Meanwhile, the frequency corresponding to *TDOS* peaks for CO_2 is always lower than that for CH_4, indicating that CO_2 has a stronger interaction with $BmimBF_4$, which is proved by the lower E_{int} between CO_2 and $BmimBF_4$ (**Figure S7**). Such huge differences in distribution, HB network, and vibrational spectrum of confined CO_2 and CH_4 imply that GO-SILMs can significantly impact their transportation.

3.2. *Solvation and diffusion behaviors of gases in GO-SILMs*

The regulated interacting strength and HB network of gases in GO-SILMs can further affect the solvation and diffusion capacity of confined CO_2 and CH_4. As mentioned above, the solvation process of gases into a GO-SILM can be described by ΔG_{sol} (Fig. 4**a**). Fig. 4**b** displays the representative ΔG_λ of CH_4 as a function of λ in bulk and confined $BmimBF_4$ at 300 K and 1 atm. Summation over all intermediate states ranging from the decoupling to coupling leads to the final ΔG_{sol} (Fig. 4**c**). The ΔG_λ first increases, then decreases with λ, reaching the maximum at $\lambda \sim 0.2$, which appears that $\Delta\lambda = 0.005$ is sufficiently fine to recover the LJ and coulombic interactions properly [38]. For gases in bulk $BmimBF_4$, ΔG_{sol} of CO_2 and CH_4 are −1.27 and 7.44 kJ/mol, respectively, showing the high CO_2 affinity of $BmimBF_4$. With $BmimBF_4$ confined in BLGOs, for both CO_2 and CH_4 in GO-SILMs, ΔG_{sol} will decrease first and then increase as *d* ranges from 1.25 to 4.00 nm, and ΔG_{sol} arrives at the minimum at $d_C = 1.50$ nm. To be exact, ΔG_{sol} of CO_2 and CH_4 in the GO-SILM with $d = d_C$ is respectively −14.64 and −1.30 kJ/mol, showing a reduction of 1052.76 % and 117.47 % for gases in bulk $BmimBF_4$, demonstrating that BLGOs enhance the affinity of $BmimBF_4$ to CO_2 and CH_4.

To quantitatively compare the solubility of CO_2 and CH_4 in GO-SILMs, the effective Henry's law constant (K_{Hr}) is defined based on ΔG_{sol}:

$$K_{Hr} = RT\frac{\rho_{IL}}{M_{IL}}exp\left(\frac{\Delta G_{sol}}{RT}\right) \tag{2}$$

where *R* is the ideal gas constant, M_{IL} and ρ_{IL} are molecular weight and density of the confined or bulk $BmimBF_4$. As displayed in Fig. 4**d**, K_{Hr} for CO_2 and CH_4 in bulk $BmimBF_4$ is respectively 73.85 and 2425.68 atm, agreeing well with the previous works [18,48–53], which proves the rationality of simulations conducted herein. As for CO_2 and CH_4 in GO-SILMs, K_{Hr} is at least about an order of magnitude lower than gases in bulk ILs. The reductions of ΔG_{sol} and K_{Hr} indicate that the microenvironments of GO-SILMs should favor CO_2 and CH_4 molecules solvating, which concurs with recent experiments [25,27,54]. It should be noted that both ΔG_{sol} and K_{Hr} decrease first and then increase, and finally approach the bulk value, where the critical point is $d_C = 1.50$ nm. The abnormal larger ΔG_{sol} and K_{Hr} at $d = 1.25$ nm should originate from the compact double-layer structure of confined ILs, where such double-layer ILs can provide less CO_2 interacting sites and adverse to the solvating process of CO_2 in the confined ILs. The solvating advantages of microenvironments originate from the interfacial structure reconstruction of

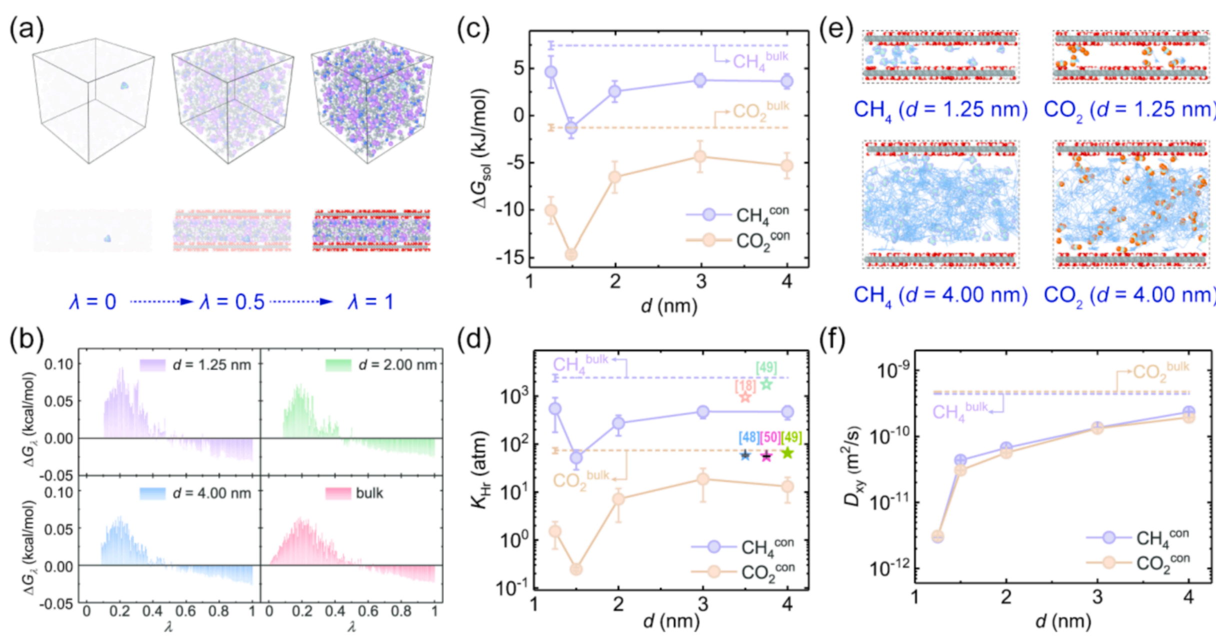

Fig. 4. Solvation and diffusion behavior of gases in GO-SILMs. (a) The atomic structures of CH_4 in bulk ILs (upper) and the GO-SILM (lower), at $\lambda = 0$, 0.5, and 1. (b) The ΔG_λ as a function of λ, for CH_4 in bulk and confined $BmimBF_4$ at $T = 300$ K and $p = 1$ atm. (c-d) The ΔG_{sol} and K_{Hr} of CO_2 and CH_4 in GO-SILMs change with d, referenced to gases in bulk ILs, respectively. The solid and hollow pentagrams represent K_{Hr} of CO_2 and CH_4 in bulk $BmimBF_4$, reported in recent experimental and simulation works of literature [18,48–50], as shown in the label. (e) The atomic structures and trajectory lines of gases in GO-SILMs with $d = 1.25$ and 4.00 nm, where the blue represents the trajectory line, and $BmimBF_4$ was not shown for clarity. (f) The D_{xy} of CO_2 and CH_4 in GO-SILMs changes with d, referenced to gases in bulk ILs, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

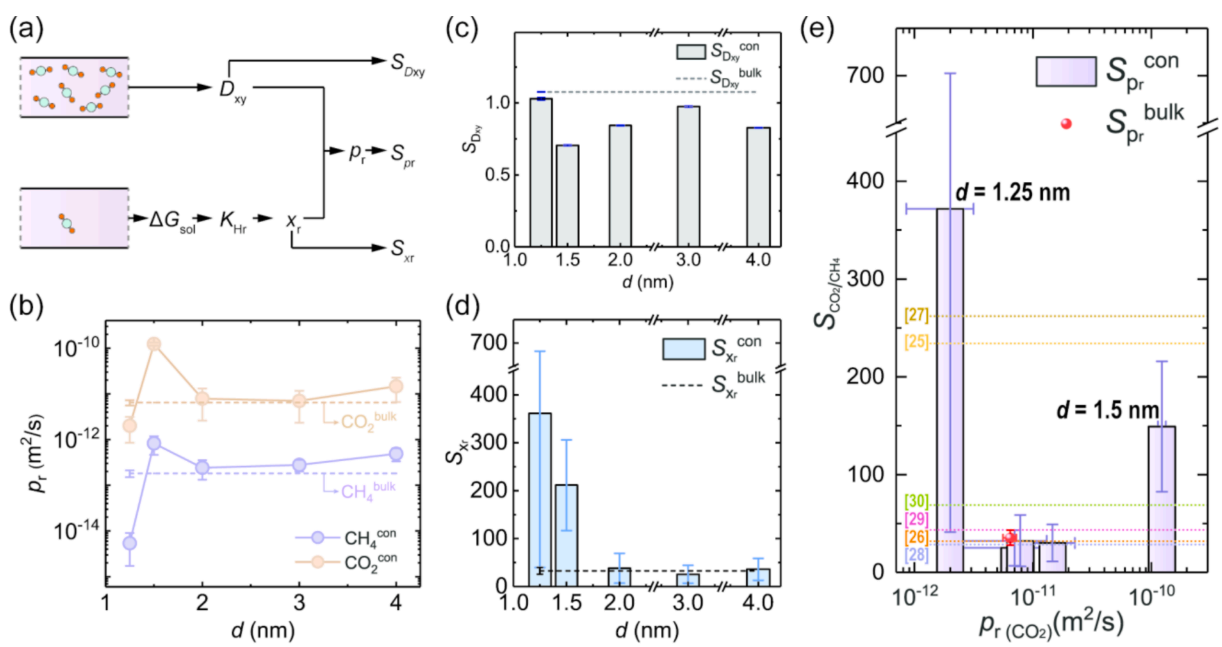

Fig. 5. Effective separating performance of CO_2/CH_4 in GO-SILMs. (a) The flow chart for the selectivity calculation, where the violet box and colored molecules represent the solvent and the solute molecule. (b) The p_r of CO_2 and CH_4 in GO-SILMs changes with d, referenced to gases in bulk ILs. (c-d) The S_{Dxy} and S_{xr} change with d, compared to gases in bulk ILs, respectively. (e) The S_{pr} changes with the $p_{r(CO2)}$, compared to S_{pr} of gases in bulk ILs. The dotted lines represent the values of $S_{CO2/CH4}$ in 2D -SILMs, reported in recent experimental works of literature [25–30], as shown in the label. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$BmimBF_4$, especially in the near-wall layers. As discussed above, with the rise of d, $P_{com\text{-}B}$ of $BmimBF_4$, CO_2, and CH_4 decreases monotonously (**Figure S4**). Moreover, for each component, E_{int} of CO_2 is also invariably stronger than that of CH_4 (**Figure S7**), consistent with the higher RDF peak intensity of CO_2-BF_4^- and O_2-H (Fig. 2**c&S6**), echoing with that CO_2 possesses the lower ΔG_{sol} as well.

In addition to solvation, diffusing capacity is another essential factor in the permeability of gases. As shown in Fig. 4**e**, owing to the stronger restriction from GO walls with $d = 1.25$ nm, both CO_2 and CH_4 molecules in this GO-SILM can only activity within a very localized space. As d rises to 4.00 nm, the footprint of gas molecules will be distributed in the entire region of BLGOs and rarely enter the near-wall layers. Such

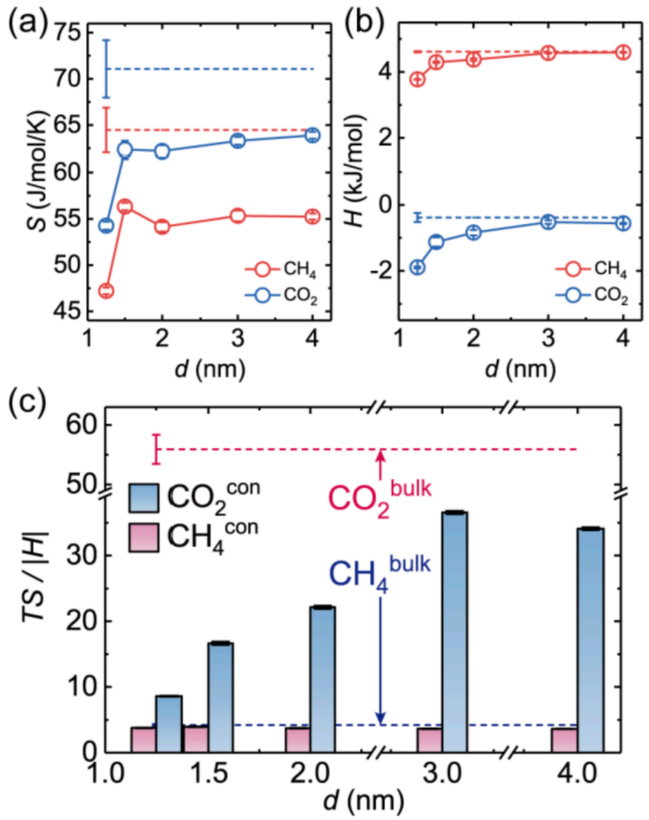

Fig. 6. Thermodynamic analysis of gases in GO-SILMs. (a-b) The S and H of CO_2 and CH_4 in GO-SILMs change with d, referenced to gases in bulk ILs, respectively. (c) The $TS/|H|$ of CO_2 and CH_4 in GO-SILMs changes with d, referenced to gases in bulk ILs, respectively.

Table 1
ΔG_{sol} of gases in a BLGO with different conditions. (Unit: kJ/mol).

ΔG_{sol}	Filler			Solvent	
	None	Water	$BmimBF_4$	GO-SILMs	$BmimBF_4$
CO_2	945.89 ± 0.00	6.87 ± 0.39	−14.64 ± 0.19	−14.64 ± 0.19	−1.67 ± 0.10
CH_4	1132.90 ± 0.05	15.75 ± 1.02	−1.30 ± 1.10	−1.30 ± 1.10	7.65 ± 0.10

confinements can greatly restrain the D_{xy} of CO_2 and CH_4. As displayed in Fig. 4**f**, D_{xy} of CO_2 and CH_4 in GO-SILMs decreases in the order of magnitude compared with that in bulk ILs. With d growing from 1.25 nm to d_C, the expansion of nanoconfined space can promote the D_{xy} of gases from 0.30 to 4.36 × 10^{-11} m^2/s (Fig. 4**e-f&2a**). As d continuously expands, the D_{xy} of gases will rise and reach the same order of magnitude as the bulk ILs at d = 3.00 nm. The alterant ΔG_{sol} and D_{xy} of CO_2 and CH_4 in GO-SILMs suggest that the strategy of nanoconfinement should be an effective method to tune the $S_{CO2/CH4}$ as presented in the recent experiments [25,27,28,54], whose inner mechanism will be analyzed in the following section.

3.3. The enhanced CO_2 separating performance of GO-SILMs

Following the flow chart in Fig. 5**a**, the effective permeability coefficient (p_r) is calculated as $p_r = D_{xy} \cdot x_r = D_{xy} \cdot p/K_{Hr}$, where x_r is the effective solubility. The relationship between p_r and d for CO_2 and CH_4 in GO-SILMs is summarized in Fig. 5**b**, showing that the p_r of CO_2 is far larger than that of CH_4. Meanwhile, p_r of both CO_2 and CH_4 rises first and then declines with d, peaking at $d = d_C$, agreeing well with the evolution of ΔG_{sol}-d and K_{Hr}-d (Fig. 4**c-d**). For instance, p_r for CO_2 and CH_4 in the GO-SILM with $d = d_C$ is 1.23 × 10^{-10} and 8.26 × 10^{-13} m^2/s, respectively 19.17 and 4.55 times of corresponding values in bulk ILs. Regardless of d = 1.25 nm, p_r for both CO_2 and CH_4 in GO-SILMs is higher than gases in bulk ILs, proving that the microenvironments of GO-SILMs can promote gas transportation in ILs.

The separating performance can be described by the effective selectivity of CO_2 over CH_4 (S_{pr}), composed of diffusive selectivity (S_{Dxy}) and solubility selectivity (S_{xr}), expressed as:

$$S_{p_r} = S_{D_{xy}} \cdot S_{x_r} = \frac{D_{xy}(CO_2)}{D_{xy}(CH_4)} \cdot \frac{x_r(CO_2)}{x_r(CH_4)} \tag{3}$$

combined with x_r and D_{xy}, the evolution of S_{Dxy}, S_{xr}, and S_{pr} with varying d is summarized in Fig. 5**c-d&S8**. As shown in Fig. 5**c**, S_{Dxy} slightly fluctuates around 1, reflecting that the diffusion selectivity is inapparent. GO-SILMs even show disadvantageous S_{Dxy} compared with bulk ILs, similar to the diffusivity selectivity reported recently [55]. By comparing S_{xr} and S_{pr}, the fact can be found that whether CO_2 and CH_4 in GO-SILMs or bulk ILs, S_{xr}, and S_{pr} are always very close, showing the dominant role of S_{xr} in S_{pr} [55]. The minimum S_{pr} is 25.01 and smaller than S_{xr} owing to the $S_{Dxy} < 1$, confirming that $BmimBF_4$ has the excellent selectivity of CO_2 over CH_4. As shown in Fig. 5**e**, the S_{pr} and p_r of CO_2 [$p_{r(CO2)}$] for the various systems are summarized, indicating that the greatest S_{pr} and the maximum $p_{r(CO2)}$ cannot be satisfied simultaneously. Meanwhile, the S_{pr} for GO-SILM with d = 1.25 to 4.0 nm ranges from 25.01 to 371.82, which agrees well with the reported experiment result and reflects the rationality of simulation conducted here. On the bright side, **Figure S8** shows that S_{xr} of the GO-SILM with d = 1.25 and 1.50 nm is 361.18 and 211.46, respectively, larger than recent experimental results. The higher selectivity of GO-SILM with d = 1.25 nm should originate from the stronger CO_2-ILs [56,57] and CO_2-GO interactions, and the double-layer confined structure of ILs. To put it another way, confined ILs should be a viable and optional way to break through the bottleneck of membrane separation selectivity and permeability competition.

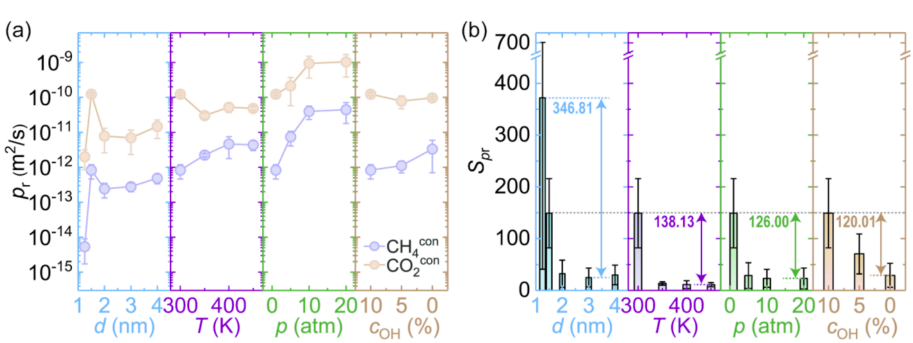

Fig. 7. Effective separating performance of CO_2/CH_4 in GO-SILMs on different conditions. (a-b) The p_r and S_{pr} of gases in GO-SILMs change with d, T, p, and c_{OH}, respectively.

3.4. Thermodynamical origin of the enhanced separating ability for GO-SILMs

To reveal the mechanism of interlayer distance on thermodynamic properties of CO_2 and CH_4 in GO-SILMs, the S, H, and A per gas molecule are further analyzed, which can be expressed as a sum of vibrational, translational, and rotational components (**Figure S9**). As shown in Fig. 6**a-b&S10**, CO_2 and CH_4 in GO-SILMs have lower S, higher A, and lower H than gases in bulk ILs. Once CO_2 or CH_4 is in the GO-SILM with $d = 1.25$ nm, almost all gas molecules are in the near-wall layers and bear the stronger restriction, possessing the lowest S and the more N_{HB} (Fig. 3**b**). As d rises from 1.25 to 4.00 nm, the S of both CO_2 and CH_4 changes non-monotonically, showing a maximum value at $d = d_C$. The extreme value of S demonstrates that gases in the GO-SILM with $d = d_C$ can lose little S relative to gases in bulk ILs, which leads to the maximum x_r and further causes the maximum p_r for both CO_2 and CH_4. Similarly, the evolution of A-d for gases shows a minimum value at $d = d_C$ (**Figure S10**), implying that when $d \leq 2.00$ nm, the GO-SILM with $d = d_C$ should be the easier confinement for gases to enter, consistent with S-d. Furthermore, whether gases in GO-SILMs or bulk ILs, CO_2 always has a higher S and a lower A than CH_4, proving it is easier for CO_2 to enter $BmimBF_4$ than CH_4, which brings about the huge difference of x_r between CO_2 and CH_4 and further contributes to high S_{pr}.

Unlike the non-monotonical behaviors of S-d and A-d, the tendency of H-d is monotonic (Fig. 6**b**). Obviously, with d rising, H goes up until converging to the value in bulk ILs. To quantify the contribution of S and H and further determine the intrinsic leading role of high S_{pr}, the ratio of energy produced by S relative to H ($TS/|H|$) is reported in Fig. 6**c**. It can be confirmed that for both CO_2 and CH_4 in bulk ILs, the energy produced by S is far more than H. For example, $TS/|H|$ of CO_2 is 55.91 while that of CH_4 is 4.19. Likewise, the $TS/|H|$ for gases in GO-SILMs is greater than 3.60, although less than the corresponding value in bulk ILs, but is enough to prove the decisive position of S. In addition, compared with CH_4, CO_2 in ILs always has a larger $TS/|H|$. Thus, for CO_2 and CH_4 in GO-SILMs or bulk ILs, the energy contribution of S is far larger than H, confirming that S drives significant amounts of CO_2 and CH_4 dissolved into ILs, where the dissolved amounts of CO_2 are much more than those of CH_4, further causing high S_{pr} of GO-SILMs.

Physically, the high S_{pr} of GO-SILMs should originate from the huge difference of x_r between CO_2 and CH_4, driven by S and enhanced by selecting the species of ILs and 2D nanosheets. To quantify the function of $BmimBF_4$ in gases dissolution into GO-SILMs, we summarized ΔG_{sol} of CO_2 and CH_4 in the BLGO filled with $BmimBF_4$ ($d = d_c$), filled with water ($d = 0.87$ nm) [25], and no filler ($d = 0.46$ nm) in Table 1. Taking CO_2 in the BLGO with no filler as an example, where the ΔG_{sol} is 945.89 kJ/mol, as for the BLGO filled with water, ΔG_{sol} reduces to 6.87 kJ/mol, when replacing water with $BmimBF_4$, ΔG_{sol} further debases to -14.64 kJ/mol. Hence, ΔG_{sol} of CO_2 in the BLGO can be reduced by at least 960.53 kJ/mol due to the filling of $BmimBF_4$. Furthermore, compared with considering GO-SILMs as the solvent, ΔG_{sol} of only regarding $BmimBF_4$ as the solvent will grow by 12.97 kJ/mol at most (Table 1), confirming that BLGOs can help CO_2 and CH_4 obtain greater x_r in $BmimBF_4$, however, the contribution is lower than filling $BmimBF_4$. In other words, $BmimBF_4$ plays a more important role in the dissolution of gases into GO-SILMs.

3.5. Additional discussion

The factors like d, T, p, c_{OH} can be easily changed in realistic applications. To obtain the most effective method for improving $S_{CO2/CH4}$, the p_r and S_{pr} of systems with different conditions are summarized in Fig. 7**a-b**. In the condition of $T = 300$ K, $p = 1$ atm, and $c_{OH} = 10\%$, with d decreasing from 4.00 to 1.25 nm, CO_2 and CH_4 in GO-SILMs show a maximum p_r at $d = d_C$, and display a maximum S_{pr} at $d = 1.25$ nm, where the promotion of S_{pr} is even high up to 346.81. Assuming $d = d_C$, in the conditions of T cools from 450 to 300 K, p reduces from 20 to 1 atm, and c_{OH} increases from 0 to 10%, S_{pr} will respectively go up by 138.13, 126.00, and 120.01, which are both far lower than the enhancement originated from changing d. Thus, compared with changing T, p, or c_{OH}, choosing a suitable d should be the most effective method for improving S_{pr} of GO-SILMs. However, the highest S_{pr} and the maximum $p_{r(CO2)}$ still cannot be obtained at the same time, for instance, with p reducing from 20 to 1 atm, $p_{r(CO2)}$ abates while S_{pr} rises.

Considering the solubility dominates S_{pr} of GO-SILMs [6], ΔG_{sol} and x_r of CO_2 and CH_4 in GO-SILMs with different d, T, p, and c_{OH} are also further studied (**Figure S11**). Compared with d and T, the effect of p and c_{OH} on ΔG_{sol} is insignificant. For example, ΔG_{sol} of CO_2 goes up by 8.19 kJ/mol with T heating up from 300 to 450 K, which only at most rises by 2.59 kJ/mol by changing p or c_{OH}. The corresponding d is further summarized in Table S1, which shows that d increases at least 0.102 nm with T heating up, while d is almost unchanged as p or c_{OH} changes. Therefore, the obvious adjustment of T to ΔG_{sol} is inseparable from the significant change of d: with T heating, d increases remarkable, the interfacial structure of GO-SILMs reconstructs, S of gases in GO-SILMs changes and further drives the rise of ΔG_{sol}. Unlike the dominance of d and T in controlling ΔG_{sol}, p is the more powerful factor affecting x_r due to $x_r = p/K_{Hr}$. In addition, **Figure S11** indicates the feasibility of improving x_r of CO_2 via adjusting d, decreasing T, rising p, or increasing c_{OH}. Combined with Fig. 7**a-b**, it can be proposed that reducing d to d_C, decreasing T to 300 K, and increasing c_{OH} to 10% will improve the S_{pr} and $p_{r(CO2)}$ simultaneously.

4. Conclusion

In brief, this work set out to better understand the structures, thermodynamic properties, and transportations of CO_2 and CH_4 in GO-SILMs via MD simulations. We established a series of BLGOs with d varying from 1.25 to 4.00 nm, where the ILs, CO_2, and CH_4 are confined. As d grows from 1.25 to 4.00 nm, entropy, solubility, and permeability of confined gases change non-monotonically. Meanwhile, with d expanding, gases molecules migrate from near-wall layers to the center of BLGOs, enhancing the diffusion capacity and improving the entropy, the latter one drives significant amounts of CO_2 and CH_4 dissolved into GO-SILMs, where the dissolved amounts of CO_2 are much more than those of CH_4, further causing high S_{pr}. Furthermore, compared with T, p, and c_{OH}, d is the more effective factor for regulating $S_{CO2/CH4}$ in GO-SILMs, even up to 371.82. The quantitative relationships between structures, thermodynamic properties, and transportations of CO_2 and CH_4 in GO-SILMs prove that the optimal d is an important factor for achieving a higher $S_{CO2/CH4}$, where ILs play a more important role than BLGO in the dissolution of gases in GO-SILMs, which is meaningful for the theoretical understanding, rational design, and applications of highly efficient and low-cost separation of CO_2.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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